

1. Determine the value of the indicated vector inner (dot) product $(-5, 3) \cdot (6, 8)$.
2. Determine the value of $\lim_{x \rightarrow 0} \frac{5x^3 + 3x^2 - 7x + 3}{2x - 3}$.
3. Let $\tan \theta = \frac{1}{2}$ with θ in quadrant I. Then the exact value for $\cos \theta + \cot \theta = \frac{k + w\sqrt{p}}{q}$ in simplified and reduced radical form with k , w , p , and q integers. Determine the sum $(k + w + p + q)$.
4. Determine the value of the sum $\sum_2^5 (2^k - 1)$.
5. A sine curve has a maximum value of 3 at $x = \frac{4}{3}\pi$, and a minimum value of -3 at $x = \frac{10}{3}\pi$ with exactly one zero between these two values for x . Determine the standard equation $y = A \sin[B(x - C\pi)] + D$ for this curve with x measured in radians and where $A > 0$ is the least possible value. Express your answer as the sum $(A + B + C + D)$ written as a common or improper fraction reduced to lowest terms.
6. Let $f(x) = x^3 + \frac{3}{2}$, $g(x) = 2x\sqrt{x}$, and $h(x) = \sin^2 x$. Determine the value of $f\left(g\left(h\left(\frac{\pi}{6}\right)\right)\right)$. Express your answer as a common or improper fraction reduced to lowest terms.